

Alessandro Sanvito

SOFTWARE AND AI ENGINEER

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Experience

Optiver

Amsterdam, Netherlands

SOFTWARE ENGINEER

August 2023 - Current

- Built baseline systematic strategies for new asset classes from scratch, including data sourcing, cleaning, and validation, for researchers to iterate on.
- Selected candidate signals and defined the validation criteria each baseline must meet before handoff.
- Cut backtest runtime and migrated ML model training from local to cloud with Python, Spark, Polars, DuckDB, PostgreSQL, and Delta Lake.

Mercedes-Benz (AI-SEE Project)

Stuttgart, Germany

AI RESEARCH INTERN AND MASTER THESIS STUDENT

May 2022 - June 2023

- Conducted research on 3D human avatar modeling from monocular video data in diverse real-world environments.
- Designed and implemented generative neural models, including NeRFs and Diffusion Models, to improve computer vision performance in adverse weather conditions.
- Collaborated with a cross-functional R&D team, resulting in a first-author publication at IEEE IV 2024 and a co-authored publication at ICCV 2023.

Education

KTH Royal Institute of Technology

Stockholm, Sweden

MSC. IN ICT INNOVATION, DATA SCIENCE

Sept. 2020 - May 2023

- Final grade: A - Excellent
- Implemented from scratch a diverse range of neural network architectures in NumPy, including MLPs, Hopfield Networks, and Deep Belief Networks.
- Reproduced data mining-related papers in Python and Java.

Polytechnic University of Milan

Milan, Italy

MSC. IN COMPUTER SCIENCE AND ENGINEERING

Sept. 2020 - Jul. 2023

- **1st place (academic)** at the international ACM RecSys Challenge 2021: co-developed the winning ML solution on the Twitter engagement dataset with the university team under prof. Paolo Cremonesi, using Dask, CatBoost, and XGBoost.
- Final grade: 110/110 cum laude
- Built a recommender system for a book catalogue with NumPy, SciPy, and Pandas.

Polytechnic University of Milan

Milan, Italy

BSC. IN ENGINEERING OF COMPUTING SYSTEMS

Sept. 2017 - Jul. 2020

- Final grade: 109/110
- Created a performance-oriented graph manipulation tool in C.

Publications

2024	HINT: Learning Complete Human Neural Representations from Limited Viewpoints, Sanvito Alessandro (first author), Ramazzina Andrea, Walz Stefanie, Bijelic Mario, Heide Felix ScatterNeRF: Seeing Through Fog with Physically-Based Inverse Neural Renderings,	IEEE IV 2024
2023	Ramazzina Andrea, Bijelic Mario, Walz Stefanie, Sanvito Alessandro , Scheuble Dominik, Heide Felix	ICCV 2023
2022	United We Stand, Divided We Fall: Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources, Maldini Pietro, Sanvito Alessandro , Surricchio Mattia Lightweight and Scalable Model for Tweet Engagements Predictions in a	ACM recsys'22
2021	Resource-constrained Environment, Carminati Luca, Lodigiani Giacomo, Maldini Pietro, Meta Samuele, Metaj Stiven, Pisa Arcangelo, Sanvito Alessandro , Surricchio Mattia, Maurera Fernando B. Pérez, Bernardis Cesare, Ferrari Dacrema Maurizio	ACM recsys'21

Skills

Programming	Python, SQL, Java, C++, C
Research & ML	PyTorch, generative models (NeRFs, Diffusion), computer vision, recommender systems, gradient boosting (CatBoost, XGBoost), scikit-learn
Data Engineering	Spark, Polars, DuckDB, PostgreSQL, Delta Lake, Pandas, NumPy, MLlib, cloud compute
Tools & Platforms	Git, Docker, Linux, CI/CD, pytest